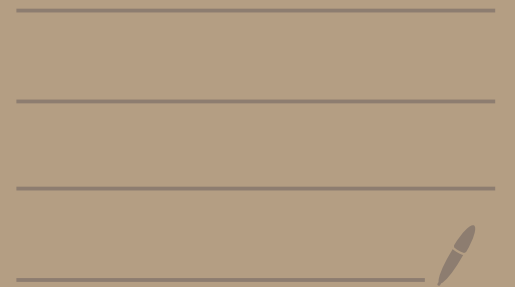
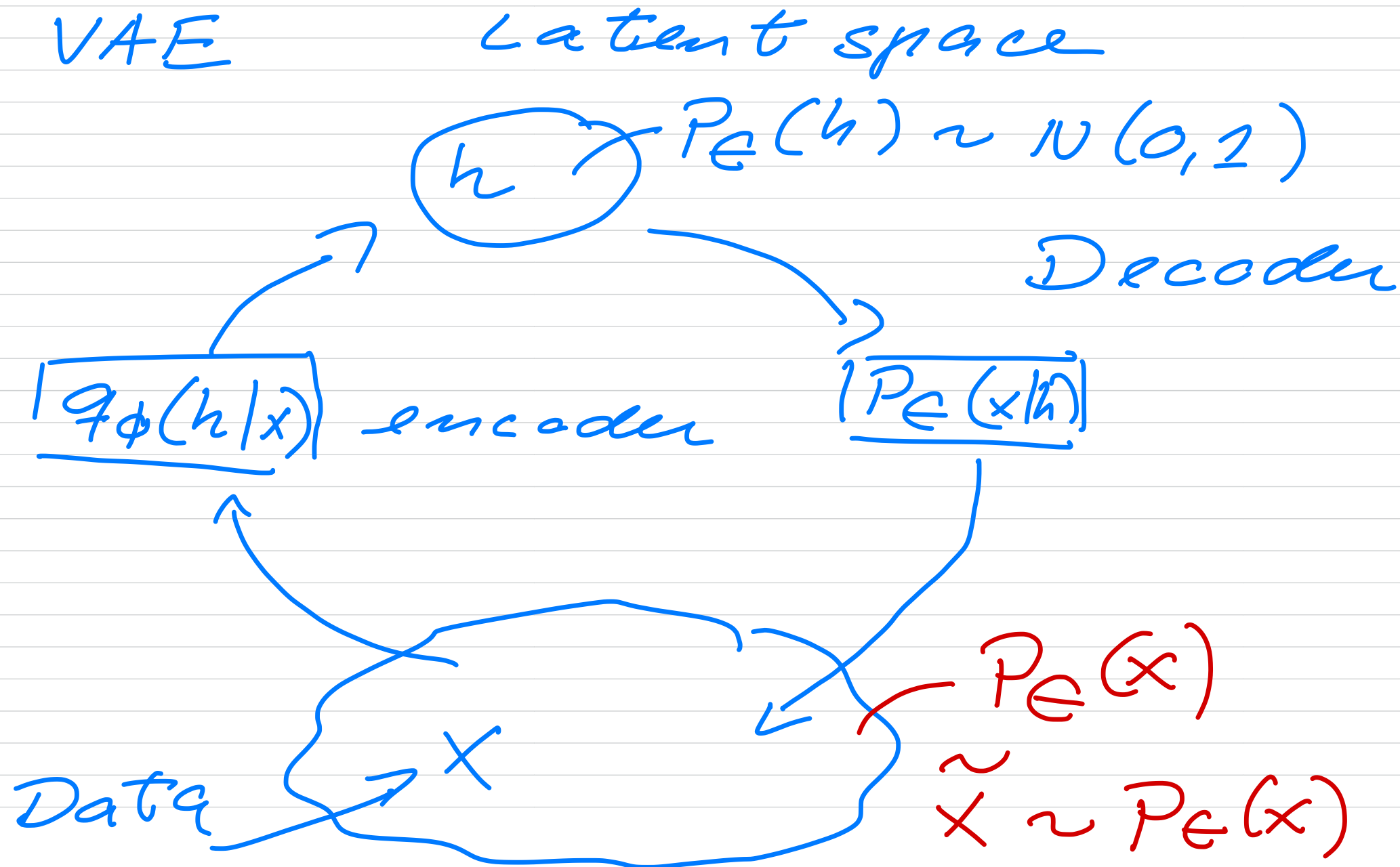


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$$P_E(x) = \int dh P_E(x|h)$$

$$= \int dh P_E(x|h) P_G(h)$$

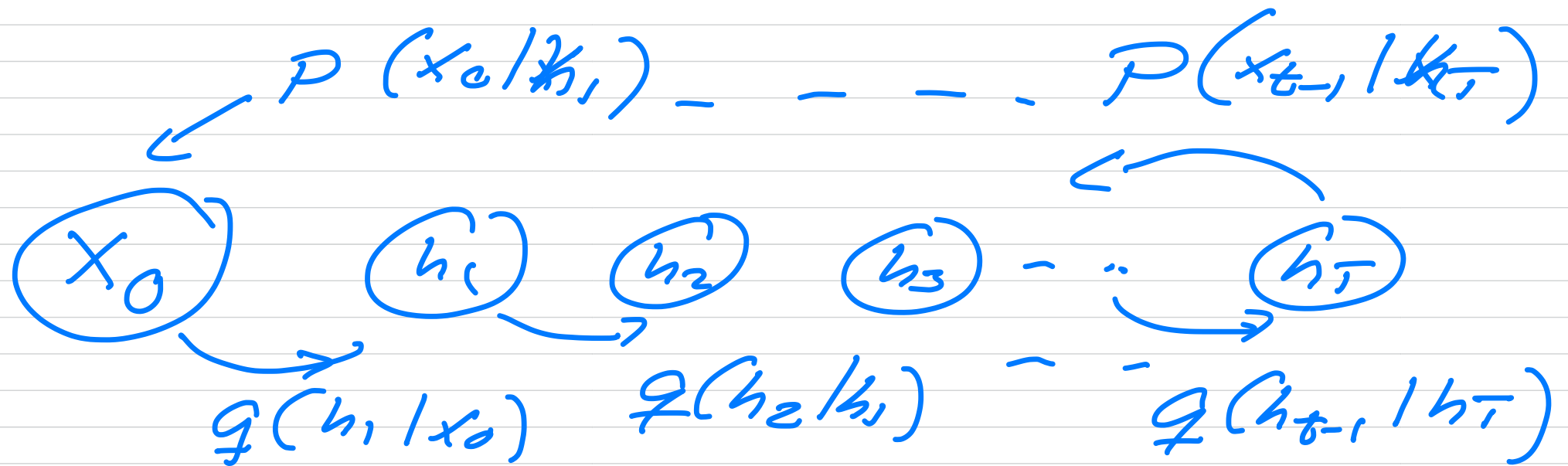
Decoder



Latent
space
(prior)



Diffusion model



Diffusion model we have
the dim of $h_1, h_2, \dots, h_t$ is
the same as x_0

$$P(x_0, h_{1:T}) = P(h_T) P_E(x_0 | h_1) \\ \times \prod_{t=2}^T P_E(h_{t-1} | h_t)$$

(Decode part with $1:T$ steps, $T \sim 100$ or more)

$$q_\phi(h_{1:T} | x_0) = q_\phi(h_1 | x_0) \\ \times \prod_{t=2}^T q_\phi(h_t | h_{t-1})$$

we optimize the marginal probability

$$\log P(x_0) = \log \int p(x_0, h_{1:T}) \times dh_{1:T}$$

$$\left(\text{VAE} \quad \log P_e(x_0) = \log \int p_e(x_0/h) p_e(h) dh \right)$$

$$= \log \int p(x_0, h_{1:T}) \frac{q_\phi(h_{1:T} | x_0) dh_{1:T}}{q_\phi(h_{1:T} | x_0)}$$

$$= \log \mathbb{E}_{q_\phi} \left[\frac{p(x_0, h_{1:T})}{q_\phi(h_{1:T} | x_0)} \right]$$

Jensen's inequality

$$\geq \mathbb{E}_{q_\phi} \left[\log \frac{p(x_0, h_{1:T})}{q_\phi(h_{1:T} | x_0)} \right]$$

$$= E_{q_\phi} \left[\frac{\log P(h_T) P_E(x_0|h_1) \prod_{t=2}^T P_\phi(h_{t-1}|h_t)}{q_\phi(h_1|x_0) \prod_{t=2}^T q_\phi(h_t|h_{t-1})} \right]$$

h_i have same dim as x_0

$$h_i \rightarrow x_i'$$

$$x_0 \rightarrow x$$

VDM = Variational Diffusion model

- Latent space = data dim
- Latent encoder is not learned, predefined linear Gaussian model

The VDM posterior

$$q(x_{1:T} | x) = \prod_{t=1}^T q(x_t | x_{t-1})$$

Encoder distributions do not depend on ϕ
we only learn the conditional probabilities

$$P_E(x_{t-1} | x_t)$$

sampling procedure

- sample Gaussian noise from $p(x_T)$
- iteratively run the denoising transitions $P_E(x_{t-1} | x_t)$

$$\log p(x) = \log \int p(x_{1:T}) dx_{1:T}$$

$$p(x_{1:T}) = p(x_1) p(x_2 | x_1) \dots p(x_T | x_{1:T-1})$$

$$= \log E_{q(x_{1:T}|x)} \left[\frac{p(x_{1:T})}{q(x_{1:T}|x)} \right]$$

$$\geq E_q \left[\log \frac{\prod_{t=1}^T p(x_t | x_{1:t-1})}{\prod_{t=1}^T q(x_t | x_{1:t-1})} \right]$$

$$\log p(x) \geq \text{Reconstruction term}$$

$$\mathbb{E}_{q(x_1|x)} [\log p_e(x|x_1)]$$

$$+ \mathbb{E}_{q(x_{T-1}, x_T|x)} \left[\log \frac{\tilde{p}(x_T)}{q(x_T|x_{T-1})} \right]$$

(no parameters to train)

Backward

$$+ \sum_{t=1}^{T-1} \mathbb{E}_{q(x_{t-1}, \dots, x_T|x)} \left[\log \frac{p_e(x_t|x_{t-1})}{q(x_t|x_{t-1})} \right]$$

Consistency term

Forward

Gaussian